torch.linalg.det#

https://pytorch.org/docs/stable/generated/torch.linalg.det.html#torch.linalg

Description:

Computes the determinant of a square matrix. Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions. torch.linalg.slogdet() computes the sign and natural logarithm of the absolute value of the determinant of square matrices. A (Tensor) – tensor of shape (*, n, n) where * is zero or more batch dimensions.

Function Signature

```
torch.linalg.det(A, *, out=None) → Tensor#
Parameters
Keyword Arguments
```

Parameters

`Keyword`

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

Code Examples

```
>>> A = torch.randn(3, 3)
>>> torch.linalg.det(A)
tensor(0.0934)

>>> A = torch.randn(3, 2, 2)
>>> torch.linalg.det(A)
tensor([1.1990, 0.4099, 0.7386])
```

Notes & Warnings

NOTE

See also torch.linalg.slogdet() computes the sign and natural logarithm of the absolute value of the determinant of square matrices.

Detailed Documentation

Documentation

Computes the determinant of a square matrix.

Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions.

torch.linalg.slogdet() computes the sign and natural logarithm of the absolute value of the determinant of square matrices.

A (Tensor) – tensor of shape (*, n, n) where * is zero or more batch dimensions.

out (Tensor, optional) – output tensor. Ignored if None. Default: None.